Amplitude–Phase Decomposition and an Energy Obstruction Mechanism for the Riemann Hypothesis

Abstract

We introduce a transport-based framework separating spectral contributions of the Riemann zeta function into phase and amplitude channels. Numerical evidence shows that phase transport recovers the spectral frequency, while amplitude growth encodes the off-critical displacement. We derive an energy lower bound showing that any nonzero amplitude growth produces a strictly positive quadratic energy density. This reduces the Riemann Hypothesis to the exclusion of such growth under spectral closure constraints.

1 Introduction

The Riemann Hypothesis (RH) asserts that all nontrivial zeros $\rho = \beta + i\gamma$ satisfy $\beta = \frac{1}{2}$.

We propose a decomposition in which:

- the phase channel encodes γ,
- the amplitude channel encodes $\beta - \frac{1}{2}$.

This identifies the obstruction mechanism with amplitude growth.

2 Model and notation

We consider a localized spectral mode of the form

$$\Psi_{\beta,\gamma}(x) = x^{\beta-\frac{1}{2}} A_\gamma(x) e^{i(\gamma \log x - \theta_\gamma)},$$

where A_γ is a slowly varying envelope.

Introduce logarithmic time:

$$\tau = \log x.$$

3 Amplitude drift decomposition

Definition 1 (Amplitude drift)**.** *Define*

$$G_{\beta,\gamma}(\tau) := \frac{d}{d\tau} \log |\Psi_{\beta,\gamma}(e^\tau)|.$$

Lemma 1 (Envelope decomposition)**.** *One has*

$$G_{\beta,\gamma}(\tau) = \beta - \frac{1}{2} + E_\gamma(\tau),$$

where

$$E_\gamma(\tau) = \frac{d}{d\tau} \log |A_\gamma(e^\tau)|.$$

Proof. Taking logarithms,

$$\log |\Psi_{\beta,\gamma}(e^\tau)| = \Big(\beta - \frac{1}{2}\Big)\tau + \log |A_\gamma(e^\tau)|.$$

Differentiation yields the result. □

4 Amplitude energy

Definition 2 (Amplitude energy)**.** *For $\tau \in \mathbb{R}$ and $L > 0$, define*

$$\mathcal{E}_{\beta,\gamma}(\tau, L) := \int_\tau^{\tau+L} |G_{\beta,\gamma}(u)|^2 \, du.$$

Lemma 2 (Boundary-form lower bound)**.** *For all τ and $L > 0$,*

$$\mathcal{E}_{\beta,\gamma}(\tau, L) \geq \Big(\beta - \frac{1}{2}\Big)^2 L + 2\Big(\beta - \frac{1}{2}\Big)\Big(\log |A_\gamma(e^{\tau+L})| - \log |A_\gamma(e^\tau)|\Big).$$

Proof. Using $G_{\beta,\gamma} = a + E_\gamma$ with $a = \beta - \frac{1}{2}$,

$$|G|^2 = a^2 + 2aE_\gamma + E_\gamma^2.$$

Integrating and using $E_\gamma^2 \geq 0$ yields the result. □

5 Amplitude-growth obstruction

Proposition 1 (Energy obstruction)**.** *Assume that the envelope satisfies the sublinear boundary condition*

$$\sup_\tau \frac{\big|\log |A_\gamma(e^{\tau+L})| - \log |A_\gamma(e^\tau)|\big|}{L} \to 0 \quad (L \to \infty).$$

Then for every $\beta \neq \frac{1}{2}$ there exists $c_\beta > 0$ such that

$$\inf_\tau \frac{\mathcal{E}_{\beta,\gamma}(\tau, L)}{L} \geq c_\beta > 0$$

for all sufficiently large L.

Proof. Divide the previous inequality by L and use the sublinear boundary assumption. □

6 Numerical validation

Numerical experiments using localized spectral constructions show:

- Phase drift satisfies
$$\frac{d}{d\tau} \arg \Psi \approx \gamma,$$
- Amplitude drift satisfies
$$\mathbb{E}[G_{\beta,\gamma}] \approx \beta - \tfrac{1}{2},$$
- Linear fit yields coefficients
$$a \approx 1, \quad b \approx -\tfrac{1}{2},$$
  stable across γ and localization parameter δ_k.

7 Interpretation

Remark 1 (Phase–amplitude separation)**.** *The spectral frequency γ appears in the phase channel, while the real part $\beta - \frac{1}{2}$ appears in amplitude growth.*

Remark 2 (Reduction of RH)**.** *The Riemann Hypothesis reduces to excluding nonzero amplitude growth under spectral closure.*

8 Conclusion

We have isolated the obstruction mechanism to amplitude growth and shown that any nonzero displacement $\beta - \frac{1}{2}$ produces a strictly positive energy density under mild assumptions.

The remaining problem is to justify the envelope control and closure constraints in the full zeta setting.

9 Renormalized amplitude obstruction

We replace the raw amplitude observable by a renormalized readout in which the envelope is removed explicitly.

Definition 3 (Renormalized amplitude readout)**.** *Let*

$$\Psi_{\beta,\gamma}(x) = x^{\beta-\frac{1}{2}} A_\gamma(x) e^{i(\gamma \log x - \theta_\gamma)}.$$

In logarithmic time $\tau = \log x$, define

$$\mathcal{R}_{\beta,\gamma}(\tau) := \log|\Psi_{\beta,\gamma}(e^\tau)| - \log|A_\gamma(e^\tau)|.$$

Lemma 3 (Exact renormalized drift)**.** *One has*

$$\mathcal{R}_{\beta,\gamma}(\tau) = \Big(\beta - \frac{1}{2}\Big)\tau, \qquad \frac{d}{d\tau}\mathcal{R}_{\beta,\gamma}(\tau) = \beta - \frac{1}{2}.$$

Proof. Taking logarithms,

$$\log|\Psi_{\beta,\gamma}(e^\tau)| = \Big(\beta - \frac{1}{2}\Big)\tau + \log|A_\gamma(e^\tau)|.$$

Subtracting $\log|A_\gamma(e^\tau)|$ yields the claim. □

Definition 4 (Renormalized amplitude energy)**.** *Define*

$$\widetilde{\mathcal{E}}_{\beta,\gamma}(\tau, L) := \int_\tau^{\tau+L} \left|\frac{d}{du}\mathcal{R}_{\beta,\gamma}(u)\right|^2 du.$$

Proposition 2 (Renormalized amplitude obstruction)**.** *For every $\beta \neq \frac{1}{2}$,*

$$\widetilde{\mathcal{E}}_{\beta,\gamma}(\tau, L) = \left(\beta - \frac{1}{2}\right)^2 L.$$

In particular, the renormalized amplitude energy density is strictly positive:

$$\frac{\widetilde{\mathcal{E}}_{\beta,\gamma}(\tau, L)}{L} = \left(\beta - \frac{1}{2}\right)^2 > 0.$$

Proof. By the previous lemma,

$$\frac{d}{du}\mathcal{R}_{\beta,\gamma}(u) = \beta - \frac{1}{2}.$$

Therefore

$$\widetilde{\mathcal{E}}_{\beta,\gamma}(\tau, L) = \int_\tau^{\tau+L} \left|\beta - \frac{1}{2}\right|^2 du = \left(\beta - \frac{1}{2}\right)^2 L.$$

□

Remark 3 (Meaning for RH)**.** *This shows that once an off-critical contribution is isolated in renormalized form, its amplitude-growth channel carries a strictly positive linear energy density. The remaining task is to prove that such a channel can be extracted from the full zeta readout and is incompatible with closure.*

10 Residual matched-mode extraction

After fitting the dominant spectral core by a three-peak Lorentzian model with quadratic baseline, the residual remains non-negligible, with residual energy above the noise floor and essentially zero correlation with the fitted core channel. This indicates that the residual is not merely a fitting defect, but contains additional structured content. In the present dataset, the three-peak core fit achieves $R^2 \approx 0.95$, while the residual is still marked as non-small and its correlation with the core channel is numerically negligible. :contentReference[oaicite:0]index=0

10.1 Residual analytic signal

Let

$$r(x) := y(x) - y_{\text{fit}}(x)$$

denote the residual after subtraction of the fitted core model. We form its analytic signal

$$\mathcal{A}_r(x) := r(x) + i\,\mathcal{H}[r](x),$$

where $\mathcal{H}$ is the Hilbert transform.

From this analytic signal we define the instantaneous phase

$$\phi_r(x) := \operatorname{unwrap} \arg \mathcal{A}_r(x)$$

and the instantaneous frequency

$$\omega_r(x) := \frac{d}{dx}\phi_r(x).$$

A stabilized scan of $\omega_r(x)$ over amplitude-supported regions reveals several persistent frequency bands, with dominant candidates near

$$\omega \approx 0.75, \quad 0.86, \quad 1.18,$$

and weaker bands extending above 1.2. The mean stabilized instantaneous frequency over the good-amplitude mask is approximately 1.20, with substantial spread, indicating a multimode residual rather than pure noise.

10.2 Matched extraction in linear x-space

Because the fitted peak parameter γ in the Lorentzian core model is a linewidth parameter, not a phase-transport frequency, it is not suitable for direct demodulation. Instead, we perform matched extraction directly in linear x-space using candidate frequencies obtained from the residual instantaneous-frequency scan.

For target frequency ω_* and growth parameter g_*, define the matched extraction

$$E_{\omega_*,g_*}[r](x) := \frac{\displaystyle\int K_{\sigma_x}(x-u)\,\mathcal{A}_r(u)\,e^{-i\omega_* u}\,e^{-g_* u}\,du}{\displaystyle\int K_{\sigma_x}(x-u)\,du},$$

where

$$K_{\sigma_x}(t) = e^{-t^2/(2\sigma_x^2)}$$

is a Gaussian localization kernel.

We then evaluate the following observables in the demodulated envelope frame:

$$\frac{d}{dx}\arg E_{\omega_*,g_*}[r](x), \qquad \frac{d}{dx}\log|E_{\omega_*,g_*}[r](x)|,$$

as well as their remodulated counterparts after multiplying back by

$$e^{i\omega_* x}e^{g_* x}.$$

10.3 Robust residual mode

A local robustness scan centered near the Stage C extra channel location

$$x_0 \approx -5.93$$

shows a stable extracted mode near

$$\omega_* \approx 0.8568.$$

This mode persists as the localization width is varied over

$$\sigma_x \in \{6, 8, 10, 12\},$$

with progressively improving matched score as σ_x increases. The best candidate occurs at

$$\omega_* = 0.85683, \qquad g_* = -0.01, \qquad \sigma_x = 12,$$

for which the demodulated envelope observables are close to flat:

$$\frac{d}{dx}\arg E_{\omega_*,g_*}[r](x) \approx -7.5\times 10^{-5} \pm 2.2\times 10^{-2},$$

$$\frac{d}{dx}\log|E_{\omega_*,g_*}[r](x)| \approx 1.5\times 10^{-3} \pm 1.4\times 10^{-2}.$$

After remodulation, the phase drift returns to the target frequency:

$$\frac{d}{dx}\arg\Big(E_{\omega_*,g_*}[r](x)e^{i\omega_* x}e^{g_* x}\Big) \approx 0.85675 \pm 0.022,$$

while the remodulated amplitude drift remains small:

$$\frac{d}{dx}\log\Big|E_{\omega_*,g_*}[r](x)e^{i\omega_* x}e^{g_* x}\Big| \approx -8.5\times 10^{-3} \pm 1.4\times 10^{-2}.$$

Moreover, the best score decreases monotonically across the tested kernel widths, from roughly 0.43 at $\sigma_x = 6$ to 0.076 at $\sigma_x = 12$, which supports the interpretation that this is a genuine locally extractable residual channel rather than a kernel artifact. The Stage C fit had already identified an additional residual structure outside the dominant three-peak core, and the matched extraction is consistent with that observation, though it operates in a different parameterization.

10.4 Interpretation

The main conclusion is not that an off-critical zeta zero has been identified. That would be too strong. What the present analysis supports is the following:

1. The dominant three-peak core does not exhaust the structured content of the spectrum.
2. The residual contains a robust, locally extractable oscillatory channel.
3. This channel is stable under matched extraction and Gaussian-width variation in linear x-space.

Thus the residual is not adequately described as a noise floor. It carries its own coherent spectral information, separated from the fitted core channel.

Remark 4. *At the current stage, the residual matched mode should be interpreted as an empirical extra channel, not yet as a rigorously identified arithmetic or zeta-theoretic mode. The analytic problem that remains is to connect such a residual channel to a principled decomposition of the full spectral readout.*

11 Residual matched-mode extraction

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12 Residual Mode Extraction and Coherent Structure

12.1 Matched Residual Extraction

Let $r(x)$ denote the residual signal obtained after subtracting the dominant core-field model from the observed data.

We define the matched extraction operator for frequency ω and growth parameter g as

$$E_{\omega,g}[r](x) := \int K_\sigma(x-y)\, r(y)\, e^{-i(\omega y + g y)}\, dy,$$

where K_σ is a localized smoothing kernel of width σ_x.

The demodulated envelope is given by

$$E_{\omega,g}[r](x),$$

while the reconstructed (remodulated) mode is

$$r_{\omega,g}(x) = \Re\Big(E_{\omega,g}[r](x)\, e^{i(\omega x + g x)}\Big).$$

—

12.2 Empirical Coherent Residual Mode

Theorem 1 (Empirical Residual Coherence)**.** *There exists a frequency ω_* and growth parameter g_* such that the matched extraction of the residual exhibits approximate phase and amplitude stationarity:*

$$\partial_x \arg E_{\omega_*,g_*}[r](x) \approx 0, \qquad \partial_x \log |E_{\omega_*,g_*}[r](x)| \approx 0,$$

over a nontrivial interval of x.

Numerically, this occurs for

$$\omega_* \approx 0.85683, \qquad g_* \approx -0.01 \ \textit{to}\ 0,$$

with stability improving as the kernel width σ_x increases.

Furthermore, remodulation recovers a coherent oscillatory component:

$$\partial_x \arg\left(E_{\omega_*,g_*}[r](x) e^{i\omega_* x}\right) \approx \omega_*.$$

—

12.3 Numerical Evidence

The following table summarizes the best matched parameters across kernel scales σ_x:

σ_x	ω_{best}	g_{best}	Score
6	0.86849	-0.01000	0.43487
8	0.86849	-0.02000	0.23644
10	0.86849	-0.01000	0.14164
12	0.85683	-0.01000	0.07597

As σ_x increases, the score decreases monotonically, indicating improved stationarity of the extracted mode.

—

12.4 Interpretation

The residual signal $r(x)$ contains a coherent oscillatory component that is not captured by the primary core-field model.

This component:

- is stable under variation of the smoothing scale σ_x,
- exhibits near-zero phase drift after demodulation,
- and reconstructs a consistent frequency upon remodulation.

We interpret this as evidence that the residual is not purely noise, but contains structured spectral content accessible via matched extraction.

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12.5 Remark

This result does not constitute a proof of any hypothesis regarding the global spectral structure (e.g., the Riemann Hypothesis), but demonstrates that the current core-field approximation is incomplete and leaves behind a measurable, coherent residual mode.

12.6 Mini-Lemma: Eulerian Lossless Readout Rotation

Let

$$\Pi : \mathbb{R}^2 \to \mathbb{C}, \qquad \Pi(x, y) = x + iy$$

be the canonical identification of the real plane with the complex field.

Let $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be the real rotation by angle θ, defined by

$$R_\theta(x, y) = (x \cos\theta - y \sin\theta,\ x \sin\theta + y \cos\theta).$$

Then the following intertwining relation holds:

$$\Pi \circ R_\theta = M_{e^{i\theta}} \circ \Pi,$$

where

$$M_{e^{i\theta}}(z) = e^{i\theta} z.$$

Consequences.

1. Real rotations correspond to phase multiplication in the complex readout:

$$R_\theta \ \longleftrightarrow\ z \mapsto e^{i\theta} z.$$

2. The readout Π is lossless (bijective), hence the transformation preserves all information.

3. Rotations in the real domain are therefore not destructive operations, but correspond to phase transport in the readout domain.